Step 2b - H10 - Higher capacity mineral drying

Tech baseline

Technologies used in the market

Existing rotary/fluid-bed/flash dryer (the asset to boost)

How it works: Fossil-fired drum, fluid bed or flash tube whose throughput is set by evaporation duty (kJ per kg water) and residence time.

How well it performs: ~1.0-2.5 kWh_th/kg water; single lines 20-100+ t/h; dryers routinely the plant bottleneck when inlet moisture or demand rises.

Limits: Capacity is thermally capped: more feed needs more burner + air handling + permit headroom; drum length/diameter fixes residence time.

Upstream mechanical dewatering (filter press, centrifuge, screen)

How it works: Remove free water mechanically before thermal drying — cheapest water removal (~1/10 the energy of thermal).

How well it performs: Cuts filter-cake moisture to ~8-15% for concentrates; classic first debottleneck step.

Limits: Cannot reach low residual moisture; not applicable to already-granular/free-draining ores and sands that enter the dryer at 5-12%.

Parallel/second dryer line (conventional debottleneck)

How it works: Install an additional rotary or fluid-bed dryer alongside the existing one.

How well it performs: Hatch magnesium project: tripled fluid-bed dryer count to go 48->100 ktpa. New large rotary line ~$8-15M installed, 12-18-month lead.

Limits: Full capex, footprint, ducting/fans/baghouse duplicated, combustion permit and ETS exposure grow; outage and tie-in complexity.

Heat-recovery / indirect-heat retrofit

How it works: Exhaust-gas heat exchangers, insulation, burner upgrades, conversion to indirect heating to free thermal margin.

How well it performs: EU cement/mineral retrofits report ~18-25% fuel reduction; recovers capacity margin indirectly.

Limits: Raises throughput only modestly (~5-15%); engineering downtime; still fossil-fired.

Inline booster drying sections (IR/hot-air) — proven in paper

How it works: Compact high-intensity IR or air dryers inserted at bottleneck points of an existing line to add evaporation without rebuild.

How well it performs: Ircon Solaronics references: removes line bottlenecks, lifts machine speed/capacity without full rebuilds; standard practice in paper/board.

Limits: Surface heating suits webs/sheets, not deep mineral beds; mineral analog largely unproven.

Microwave pre-dryer / booster (ProSeed)

How it works: 915/2450 MHz tunnel upstream (or interstage) of the existing dryer strips the constant-rate moisture load volumetrically; the incumbent dryer only finishes to spec.

How well it performs: Food analog MIP-9 boost ovens: +25% throughput at 75-300 kW per tunnel; mineral pilots show MW drying 6-13x faster and -30-50% energy vs convective, so a few-hundred-kW to ~1 MW booster can offload ~20-40% of evaporation duty.

Limits: Abrasive, high-temperature mineral belt/applicator design unproven at scale; dielectric response varies by matrix (silica poor, iron ore good); adds ~kWh_el operating cost vs gas unless efficiency + carbon economics compensate.

Why our solution can be better

A microwave booster adds evaporation capacity in meters of tunnel, not tens of meters of drum: volumetric water-selective heating removes part of the load upstream so the existing dryer finishes faster — the same +25-30% throughput effect proven by MW boost ovens in food and IR boosters in paper. It installs in a short outage at a fraction of the $8-15M and 12-18 months of a parallel line, adds no burner so no permit or ETS expansion, and its instant controllability also fixes moisture-uniformity problems that often are the true throughput limiter. For capacity-constrained plants the value is avoided capex plus earlier tonnes, not energy savings per se.

Key assumptions

• Drying is frequently the bottleneck when feed moisture or demand rises; upstream boosting gives +20-30% line throughput — proven for MW in food (MIP-9, measured) and IR in paper (measured); transfer to mineral beds assumed.

• Offloading ~2-4 percentage points of inlet moisture upstream removes ~20-40% of the existing dryer's evaporation duty (assumed; depends on inlet moisture profile).

• New parallel rotary line: ~$8-15M installed, 12-18-month lead (measured, market/capex references); a MW booster line est. <$1-3M per ~1 MW installed (assumed).

• Mineral duty (abrasion, 100-300C, dust) requires new applicator/belt engineering — the main technical risk, plus dielectric-loss variability by mineral (measured spread 40-78% drying efficiency).

• Booster opex: ~0.7-1.0 kWh_el/kg water removed vs ~1.2-2.0 kWh_th/kg in the incumbent; at EU industrial power ~EUR60-100/MWh vs gas ~EUR30-50/MWh, opex is roughly neutral-to-better while throughput is the payback driver (assumed).

Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input
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Decision	Decision	Go to next step	Go to next step	Go to next step	Rework	Rework	Rework	Drop the hypothesis	Drop the hypothesis	Drop the hypothesis